

- 13** The rate of removal of aspirin, a pain–killer drug, from the body follows a first–order reaction. It takes 6.0 hours for 87.5% of aspirin to be removed from the body.

What is the value of the rate constant for this reaction?

A 0.462 h^{-1}

B 0.347 h^{-1}

C 0.116 h^{-1}

D 0.0385 h^{-1}